

ED-SSS (Event Driven Smart Surveillance System)

Project Report

Team Members

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Abstract

Modern surveillance systems continuously process every video frame, regardless of whether any meaningful activity is occurring. This leads to excessive computational load, unnecessary power consumption, large storage requirements, and inefficient resource utilization.

ED-SSS (Event Driven Smart Surveillance System) is a neuromorphic-inspired surveillance platform designed to process only meaningful events instead of continuously processing all frames. Inspired by the working principles of biological neurons and event-based vision systems, the proposed solution detects motion-based changes, converts them into event spikes, performs adaptive analysis, and generates intelligent surveillance insights.

The system includes live camera monitoring, event detection, adaptive threshold learning, spike generation, threat assessment, event heatmap generation, dynamic power scaling, and a neuromorphic memory layer. By focusing only on significant visual changes, ED-SSS reduces unnecessary processing while improving surveillance efficiency.

1. Introduction

Traditional CCTV systems operate on a frame-based architecture. Every frame captured by the camera is processed, analyzed, stored, and transmitted regardless of its relevance.

For example, if a camera monitors an empty corridor for several hours, thousands of identical frames are still processed. This creates three major problems:

1. High computational cost
2. High storage requirements
3. High power consumption

Biological brains do not work in this manner. Human neurons remain mostly inactive and fire only when meaningful stimuli are detected.

This observation inspired the development of ED-SSS.

The project attempts to mimic this behavior by processing only meaningful visual events instead of every frame.

2. Problem Statement

Current surveillance systems suffer from:

- Continuous frame processing
- Wasted computational resources
- Excessive storage consumption
- High power requirements
- Difficulty identifying meaningful activity

The challenge is to design a surveillance system that processes only relevant visual information while reducing unnecessary workload.

3. Motivation

During our research, we studied event-driven computing and neuromorphic vision systems.

We observed that biological vision systems do not continuously process complete visual scenes. Instead, they react only when changes occur.

This inspired us to build a software-based event-driven surveillance framework capable of:

- Detecting meaningful changes
 - Ignoring redundant information
 - Simulating neuromorphic behavior
 - Improving surveillance efficiency
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4. Research Background

The project draws inspiration from:

Neuromorphic Computing

Neuromorphic systems imitate the behavior of biological neural networks.

Instead of continuously processing information, they respond only when events occur.

Event-Based Vision

Traditional cameras capture complete frames at fixed intervals.

Event cameras capture only pixel-level changes.

Although we used a standard RGB camera, we simulated event-driven processing through software.

Spike-Based Processing

In biological neural systems, neurons communicate through spikes.

Similarly, ED-SSS converts detected motion events into digital spikes.

5. Proposed Solution

The proposed system processes only meaningful visual changes.

Whenever a significant change is detected between consecutive frames:

- An event is generated
- A spike is produced
- The event is analyzed
- The surveillance dashboard is updated

This allows the system to focus on important activity while ignoring redundant information.

6. System Architecture

Input Video Stream



Frame Acquisition



Motion Analysis



Event Generation



Spike Encoding



Adaptive Learning Layer



Threat Assessment Engine



Heatmap Generation



Visualization Dashboard

Each stage performs a specific function within the event-driven surveillance pipeline.

7. Methodology

Step 1: Video Acquisition

The user uploads a video or activates the live camera stream.

The system receives visual data through OpenCV.

Step 2: Frame Comparison

Each incoming frame is converted into grayscale.

The current frame is compared with the previous frame.

A difference image is generated using absolute pixel difference calculations.

Step 3: Event Detection

The average intensity of the difference image is calculated.

If the motion score exceeds the adaptive threshold, the system identifies a meaningful event.

Step 4: Spike Generation

Every detected event generates a spike.

Spikes represent meaningful visual activity within the surveillance scene.

Step 5: Spike Burst Detection

When multiple spikes occur continuously within a short period, the system classifies them as a burst.

Spike bursts indicate intense activity or sudden movement.

Step 6: Adaptive Learning

The system contains a neuromorphic memory layer.

After every analysis:

- Average motion is calculated
- Baseline activity is updated
- Detection thresholds are adjusted

This allows the system to adapt to different environments.

Step 7: Threat Assessment

Threat levels are determined using spike density.

Threat Categories:

LOW

MEDIUM

HIGH

CRITICAL

Higher event density indicates greater activity within the monitored region.

Step 8: Heatmap Generation

Motion information is accumulated throughout the video.

The accumulated motion matrix is converted into a color heatmap.

The heatmap highlights regions with maximum activity.

8. Features Implemented

Event Driven Processing

Processes only meaningful visual changes.

Live Camera Monitoring

Real-time surveillance using webcam input.

Adaptive Threshold Learning

Thresholds evolve according to observed activity.

Spike Generation

Motion events are converted into spikes.

Spike Burst Detection

Detects rapid sequences of events.

Threat Classification

Automatically identifies activity severity.

Dynamic Power Scaling

Simulates intelligent resource utilization.

Event Heatmap

Visual representation of activity hotspots.

Neuromorphic Memory Layer

Stores learned environmental behavior.

9. Technologies Used

Frontend:

- React.js
- CSS
- Axios

Backend:

- FastAPI
- Python

Computer Vision:

- OpenCV
- NumPy

Communication:

- REST APIs
- WebSockets

Development Tools:

- VS Code

- [GitHub](#)
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10. Challenges Faced

During development we faced multiple challenges:

- Accurate motion detection under varying lighting conditions
- False positives caused by camera noise
- Real-time event streaming
- Dynamic threshold selection
- Efficient heatmap generation
- Live camera integration

These challenges were addressed through adaptive thresholds, motion filtering, and asynchronous communication techniques.

11. Results

The system successfully demonstrated:

- Real-time event detection
- Live surveillance monitoring
- Event-based processing
- Spike generation
- Adaptive learning
- Threat assessment
- Event heatmap visualization

Testing showed that the system processed significantly fewer meaningful events compared to total video frames, demonstrating the effectiveness of event-driven surveillance.

12. Future Enhancements

Future versions may include:

- Spiking Neural Networks (SNN)
- FPGA acceleration

- Edge AI deployment
 - Multi-camera coordination
 - Facial recognition
 - Anomaly detection using AI
 - Integration with real event cameras
 - Dedicated neuromorphic hardware
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13. Conclusion

ED-SSS demonstrates how neuromorphic-inspired event-driven surveillance can improve the efficiency of traditional monitoring systems.

By focusing only on meaningful changes rather than continuously processing every frame, the system reduces computational overhead while maintaining effective surveillance capabilities.

The project successfully combines computer vision, adaptive learning, event-based processing, live monitoring, and intelligent threat assessment into a unified surveillance platform and provides a foundation for future neuromorphic surveillance research.